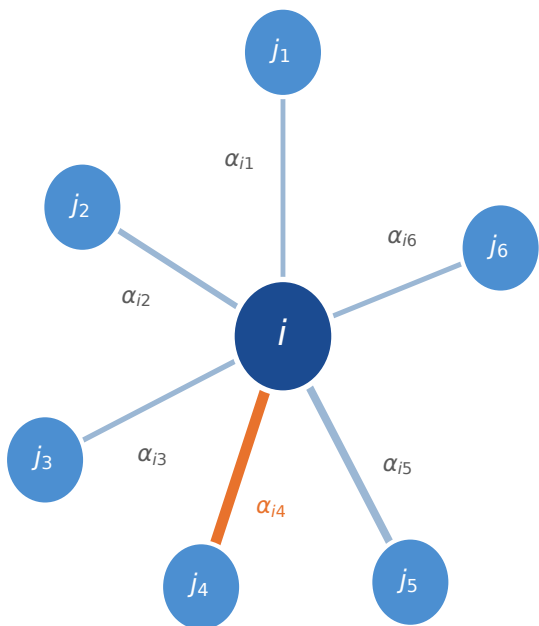


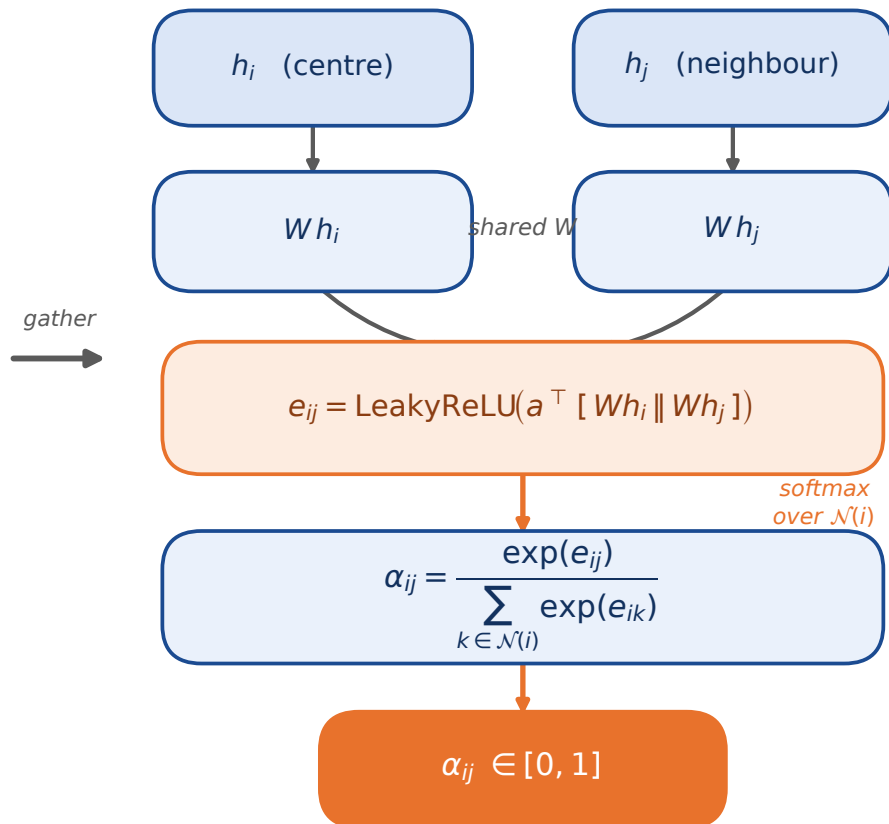
Graph Attention (GAT) layer — message passing with learned edge weights

(A) Mesh neighbourhood $\mathcal{N}(i)$

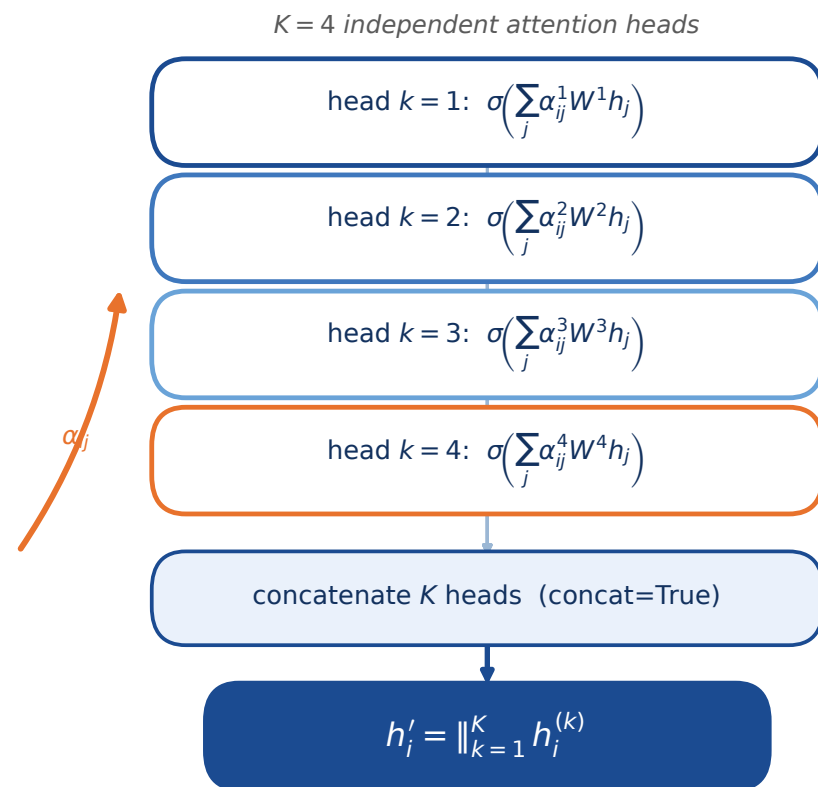


node = mesh vertex feature $h = [x, y, z, \text{DSPSS}]$
edge thickness $\propto$ learned attention α_{ij}

(B) Attention coefficient α_{ij}



(C) Multi-head update h'_i



As used in GATModel (train.py): GATConv/GATv2Conv, heads=4, concat=True, 3 stacked layers with residual projection + BatchNorm.